Program 10:

Project Structure:
Pr10
->deployment.yaml
->service.yaml

Step 1: Create deployment.yaml

```yaml
# deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

Step 2: Create service.yaml

```yaml
# service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
  type: NodePort
```

Step 3: Apply deployment & service

```
kunaltiwari@riot pr10 % kubectl apply -f deployment.yaml

deployment.apps/nginx-deployment created
kunaltiwari@riot pr10 % kubectl apply -f service.yaml

service/nginx-service created
```

Step 4: Verify resources

```
kunaltiwari@riot pr10 % kubectl get nodes

NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   10d   v1.34.0
kunaltiwari@riot pr10 % kubectl get pods

NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-7ccccd94f7-5qp8f   1/1     Running   0          60s
nginx-deployment-7ccccd94f7-gzm2v   1/1     Running   0          60s
kunaltiwari@riot pr10 % kubectl get svc

NAME            TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
kubernetes      ClusterIP   10.96.0.1      <none>        443/TCP        10d
nginx-service   NodePort    10.99.71.151   <none>        80:30868/TCP   60s
kunaltiwari@riot pr10 % kubectl get deployments

NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   2/2     2            2           74s
```

Step 5 :Describe and view details

```
kunaltiwari@riot pr10 % kubectl describe svc nginx-service

Name:                     nginx-service
Namespace:                default
Labels:                   <none>
Annotations:              <none>
Selector:                 app=nginx
Type:                     NodePort
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.99.71.151
IPs:                      10.99.71.151
Port:                     <unset>  80/TCP
TargetPort:               80/TCP
NodePort:                 <unset>  30868/TCP
Endpoints:                10.244.0.17:80,10.244.0.16:80
Session Affinity:         None
External Traffic Policy:  Cluster
Internal Traffic Policy:  Cluster
Events:                   <none>
```

Step 5 :Access the service

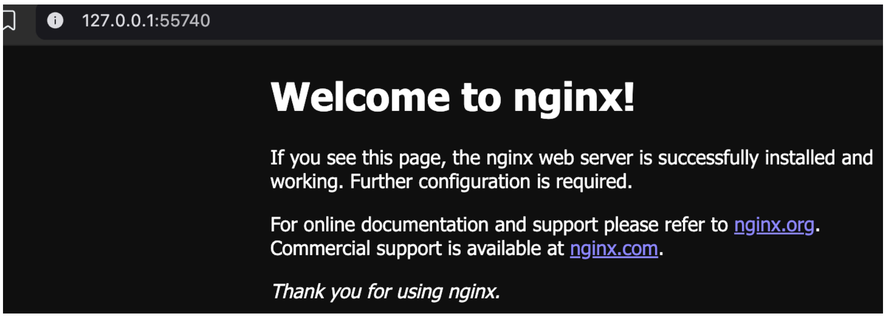
```
kunaltiwari@riot pr10 % minikube service nginx-service

| NAMESPACE |     NAME      | TARGET PORT |            URL            |
|-----------|---------------|-------------|---------------------------|
| default   | nginx-service |          80 | http://192.168.49.2:30868 |

🏃  Starting tunnel for service nginx-service.
| NAMESPACE |     NAME      | TARGET PORT |          URL           |
|-----------|---------------|-------------|------------------------|
| default   | nginx-service |             | http://127.0.0.1:55740 |

🏃  Starting tunnel for service nginx-service.
🎉  Opening service default/nginx-service in default browser...
❗  Because you are using a Docker driver on darwin, the terminal needs to 
^C✋  Stopping tunnel for service nginx-service.
```

This automatically opens the browser



Step 6 :Open Kubernetes Dashboard

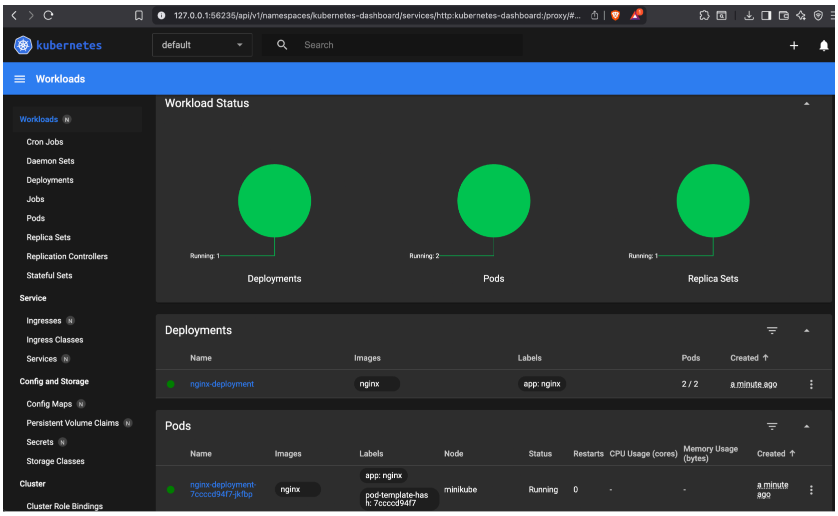
```
kunaltiwari@riot pr10 % minikube dashboard

🔌  Enabling dashboard ...
    ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
    ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
💡  Some dashboard features require the metrics-server addon. To enable all fea

        minikube addons enable metrics-server

🤔  Verifying dashboard health ...
🚀  Launching proxy ...
🤔  Verifying proxy health ...
🎉  Opening http://127.0.0.1:55747/api/v1/namespaces/kubernetes-dashboard/servi
```

This automatically opens the Kubernetes Dashboard



Step 7 :Cleanup Process

```
kunaltiwari@riot pr10 % kubectl delete svc --all

service "kubernetes" deleted from default namespace
service "nginx-service" deleted from default namespace
kunaltiwari@riot pr10 % kubectl delete deployments --all

deployment.apps "nginx-deployment" deleted from default namespace
```